R16

Code No: 138GY

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, July/August - 2022
SENSORS AND TRANSDUCERS
(Common to EEE, ECE, CSE)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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1.a) Explain the following main functions of the instruments and measurement systems:
i) Indicating function ii) Recording function iii) Controlling function
b) Draw and explain the various elements of a generalized measurement system. [6+9]

2. Explain the following types of systematic errors:
a) Instrumental errors
b) Environmental errors
c) Observational errors. [5+5+5]

3. Explain the following static characteristics:
a) Accuracy b) Sensitivity c) Drift d) Static error e) Dead zone.
[3+3+3+3+3]

4. Obtain the time response of a second order system for unit step input. [15]

5.a) Explain with the help of a diagram, the measurement of displacement using potentiometers. State the advantages and disadvantages of a potentiometer.
b) Explain the operation and construction of strain gauge. State its limitations. [8+7]

6. Explain the construction and working principle of resistance thermometer. [15]

7.a) Explain the method of measuring displacement using LVDT. State the advantages and disadvantages of LVDT.
b) Explain the working of capacitive transducer. State its applications. [8+7]

8. Write short notes on the following:
a) Digital transducers- advantages and disadvantages
b) Hall effect transducers. [8+7]

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